

FRONT END DEVELOPMENT

WEEK 10 – Fix Bugs & Refactor Code





What will you learn today?



FIX !

✓ Fix React bugs

- *By understanding the **call stack***
- *By using **breakpoints** on browser debugger*
- *By using **React DevTools** on browser*

IMPROVE !

✓ Remove non-essential state variables

- *When information is available using another state*

1 - Fix bugs by understanding the call stack

File Line number Call stack

```
Uncaught TypeError: Cannot read properties of undefined (reading 'valueEndOfYear')
    at Results (Results.jsx:7:16)
    at renderWithHooks (react-dom_client.js?v=43133a0c:11761:26)
    at updateFunctionComponent (react-dom_client.js?v=43133a0c:14149:28)
    at beginWork (react-dom_client.js?v=43133a0c:15455:22)
    at HTMLUnknownElement.callCallback2 (react-dom_client.js?v=43133a0c:3573:22)
    at Object.invokeGuardedCallbackDev (react-dom_client.js?v=43133a0c:3598:24)
    at invokeGuardedCallback (react-dom_client.js?v=43133a0c:3632:39)
    at beginWork$1 (react-dom_client.js?v=43133a0c:19258:15)
    at performUnitOfWork (react-dom_client.js?v=43133a0c:18691:20)
    at workLoopSync (react-dom_client.js?v=43133a0c:18630:13)
```

THE CONSOLE TAB

Display the **call stack** when an error occurred

<> Authored
127.0.0.1:5173

- src
 - C:/Users/LENOVO/Desktop/REAC
 - client
 - components/C:/Users/LENOVO/D
 - Header.jsx
 - Results.jsx
 - UserInput.jsx

```
1 import { calculateInvestmentResults, formatter } from
2
3 export default function Results({ input }) {
4   const results = [];
5   calculateInvestmentResults(input, results);
6   const initialInvestment =
7     results[0].valueEndOfYear -
8     results[0].interest -
9     results[0].annualInvestment;
10
11   return (
```

THE SOURCE TAB

Display the **source code** (in development mode only)

Runtime errors are displayed in red

Runtime error



10 MIN

Activity 1

- ✓ Get the code from the teacher and run it
- ✓ Clear the DURATION input
- ✓ Analyze the **call stack** and the **code** to fix the bug

Tips : what is the object related to the error? How this object is created? Updated? What could be the reason why the property cannot be read?

Year	Investment Value	Interest (Year)	Total Interest	Invested Capital
1	\$11,800	\$600	\$600	\$11,200
2	\$12,736	\$600	\$1,200	\$11,200
3	\$13,728	\$600	\$1,800	\$11,200
4	\$14,784	\$600	\$2,400	\$11,200
5	\$15,904	\$600	\$3,000	\$11,200

1 – Clear this field

2 – Check the error stack on the console

3 – Find the root reason of this bug

```
2 Uncaught TypeError: Cannot read properties of undefined (reading 'valueEndOfYear')
    at Results (Results.jsx:7:16)
    at renderWithHooks (react-dom_client.js?v=43133a0c:11761:26)
    at updateFunctionComponent (react-dom_client.js?v=43133a0c:14149:28)
    at beginWork (react-dom_client.js?v=43133a0c:15455:22)
    at HTMLUnknownElement.callCallback2 (react-dom_client.js?v=43133a0c:3573:22)
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    at workLoopSync (react-dom_client.js?v=43133a0c:18630:13)
```

```
<thead>
  <tr>
    <th>Year</th>
    <th>Investment Value</th>
    <th>Interest (Year)</th>
    <th>Total Interest</th>
    <th>Invested Capital</th>
  </tr>
</thead>
<tbody>
  {results.map((yearData) => {
    const totalInterest =
      yearData.valueEndOfYear -
```

Activity 1

Let's explore the code to understand the root cause of this bug

1

Duration field is cleared => duration = ""

duration = ""
results = []

```
export default function Results({ input }) {  
  const results = [];  
  calculateInvestmentResults(input, results);  
  const initialInvestment =  
    results[0].valueEndOfYear -  
    results[0].interest -  
    results[0].annualInvestment;  
}
```

4

This cause a bug in Results.jsx as we expect at least 1 object in the array

```
export function calculateInvestmentResults({  
  initialInvestment,  
  annualInvestment,  
  expectedReturn,  
  duration,  
}, results) {  
  let investmentValue = initialInvestment;  
  
  for (let i = 0; i < duration; i++) {  
    2
```

No loop

3

results = []

This cause the result array to be empty

We found the bug

WE will check whether the results array is empty and manage this case

2 - Fix bugs using the Browser Debugger

Breakpoints and step by step execution can help you to diagnostic the possible bugs

The screenshot shows a web browser's developer console with the 'Sources' tab open. The file 'Results.jsx' is selected, and a breakpoint is set at line 4, which contains the code `const results = [];`. The code in the editor is as follows:

```
1 import { calculateInvestmentResults, formatter } from '  
2  
3 export default function Results({ input }) {  
4   const results = [];  
5   calculateInvestmentResults(input, results);  
6   const initialInvestment =  
7     results[0].valueEndOfYear -  
8     results[0].interest -  
9     results[0].annualInvestment;  
10
```

The right-hand side of the console shows the debugger's control and watch panels. The 'Breakpoints' panel is expanded, showing the active breakpoint at line 4 of 'Results.jsx'. The 'Scope' panel is also visible, showing the current scope. The status at the bottom right indicates 'Not paused'.

Annotations with arrows point to specific features:

- 'Add a breakpoint' points to the breakpoint icon on the left margin of the code editor.
- 'Go to the next step' points to the 'Next' button (a downward arrow) in the debugger toolbar.
- 'Go inside the function' points to the 'Step Into' button (a downward arrow with a dot) in the debugger toolbar.
- 'Watch variables or expressions' points to the 'Watch' button in the right-hand panel.



Debugger works only on development mode, not with a production build



10 MIN

Activity 2

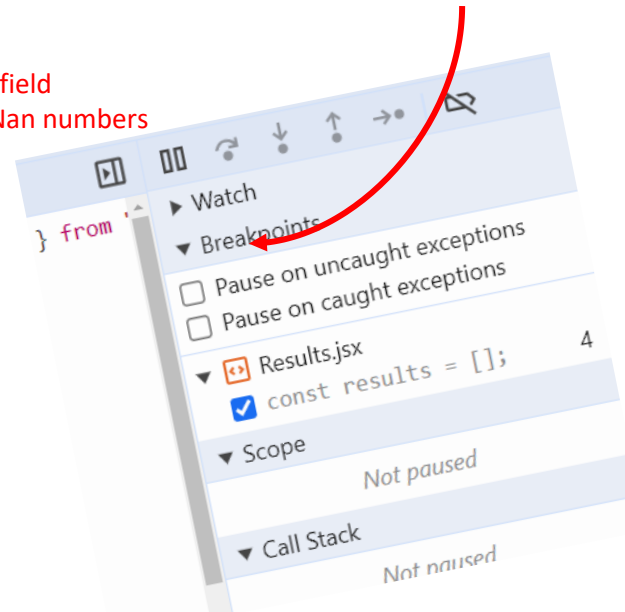
- ✓ We use the same project
- ✓ Now increment the field : Initial Investment
- ✓ You can see many rows with NaN
- ✓ Use the debugger to analyze the problem and fix the bug

Tips : what is the object related to the error? How this object is created? Updated? What could be the reason why the property cannot be read?

The image shows a web application titled "Investment Calculator". It has four input fields: "INITIAL INVESTMENT" (10000), "ANNUAL INVESTMENT" (1200), "EXPECTED RETURN" (6), and "DURATION" (10). Below the inputs is a table with columns: "Year", "Investment Value", "Interest (Year)", "Total Interest", and "Invested Capital". The first row of the table is highlighted with a red box, showing values: 1, \$11,800, \$600, \$600, and \$11,200.

- 1 – Increment this field
2- Check you see Nan numbers

2 – Use the debugger to fix the bug



3 – Find the root reason of this bug



Activity 2

Let's explore the code to understand the root cause of this bug

1

With a breakpoint on handleChange :

We could see that newValue is string, not an integer

```
11
12
13
14
15 function handleChange(inputIdentifier, newValue) {
16   setUserInput((prevUserInput) => {
17     return {
18       ...prevUserInput,
19       [inputIdentifier]: newValue,
20     };
21   });
22 }
```

"10002"

2

With a breakpoint on calculateInvestment:

The initialInvestment is still a string...

```
5 // - expectedReturn: The expected (annual) rate of
6 // - duration: The investment duration (time frame)
7 export function calculateInvestmentResults({
8   initialInvestment,
9   annualInvestment,
10  expectedReturn,
11  duration,
12 }, results) { results = []
13 let investmentValue = initialInvestment;
14
15 for (let i = 0; i < duration; i++) {
```

3

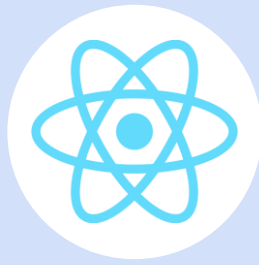
But this string is used to increment the value.. Strings don't work for additions

"4" + 4 = "44"

```
2 }, results) { results = []
3   let investmentValue = initialInvestment;
4
5   for (let i = 0; i < duration; i++) {
6     const interestEarnedInYear = investmentValue * expectedReturn;
7     investmentValue += interestEarnedInYear
8     results.push({
9       year: i + 1, // year identifier
10      interest: interestEarnedInYear // th
```

"100021800.12"

We found the bug
WE will convert the input to integer



BUG QUEST !

✓ *60 minutes to fix bugs*

✓ *Team of 2 students*



*To get points : get **your fixes validated** by the trainer*

✓ *The team with the higher score wins*



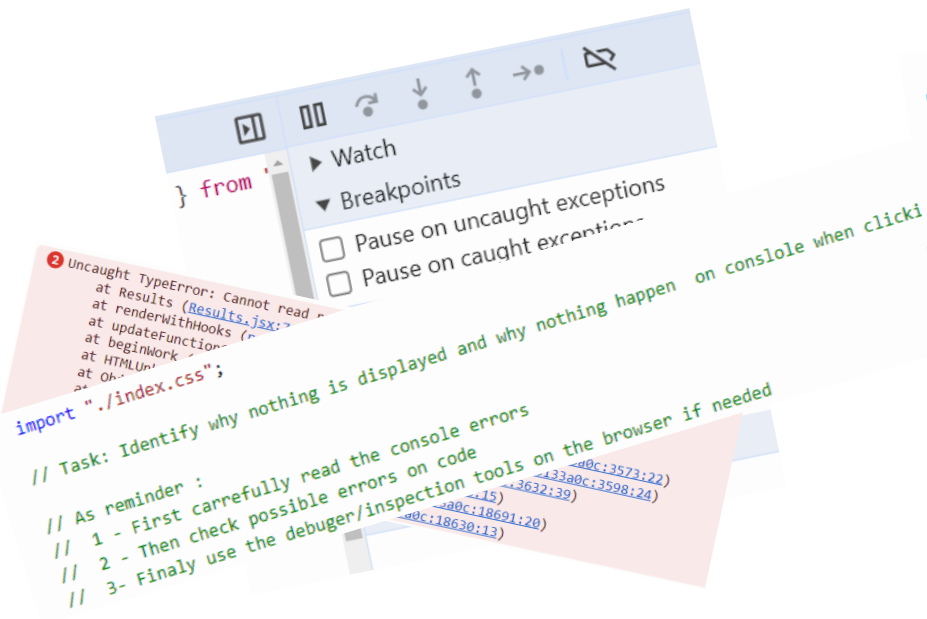
10 MIN

Activity 3

1

Fix the bugs in team

Using the **different techniques** we explained



2

Report your fixes on the game sheet

Get your fixes validated by the trainer

C2-S13 GAME			
Bug Quest!			
CHALLENGE	What was the bug?	How did you solve it?	Trainer validation
Challenge 1			
Challenge 2			
Challenge 3			

3

After 60min, the team **with the most validated fixes** win !



Go **further** after the class...

- ✓ Read Conditional Display
- ✓ Read A Component's Memory
- ✓ Read Pass function as props
- ✓ Read React Virtual DOM

- ✓ Review these slides
- ✓ Review correction code
- ✓ Update **your notes** on this course

